

10 A particle oscillates with simple harmonic motion along a line with a maximum speed v_0 .

When the displacement of the particle is half of its amplitude, its speed is

A $\frac{1}{4}v_0$.

B $\frac{1}{2}v_0$.

C $\frac{3}{4}v_0$.

D $\frac{\sqrt{3}}{2}v_0$.

11 A container of ice is heated by an electric heater. The graph below shows the variation of